

CSC 648 (Section 3)

Vibe Check: Milestone 1

ARKHA (Team 6)

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Revision	Description
v1	Created document 2/7 @5:22pm, added tech stack for Care Companion and Vibe Check apps

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Executive Summary

What's the Product?

Vibe Check is a real-time, location-based bulletin board application that helps users understand the current atmosphere, or “vibe,” of nearby places. Through short, crowd-sourced notes, users can quickly see whether a location is crowded, quiet, active, or slow. Instead of relying on outdated reviews or historical averages, Vibe Check focuses on live, user-generated updates that reflect what is happening in the moment.

What Problem Are We Solving?

Existing platforms such as Google Maps and Yelp primarily rely on historical trends and past reviews to estimate how busy or active a location might be. While useful, this information does not accurately reflect real-time conditions. As a result, people often waste time and effort traveling to places that are overcrowded, noisy, closed, or otherwise unsuitable for their immediate needs. Vibe Check addresses this gap by providing timely, location-specific updates that help users make better decisions before they arrive.

Who Is the Product For?

Vibe Check is designed for individuals who frequently make spontaneous or time-sensitive decisions about where to go. This includes university students searching for study spots, busy professionals looking for productive environments, tourists exploring unfamiliar areas, event attendees checking venue conditions, and nature enthusiasts assessing trail or park activity. The platform supports anyone who benefits from real-time situational awareness.

Value Proposition

Vibe Check enables users to avoid long waits, reduce exposure to overcrowded or noisy environments, and quickly choose the right place for their needs. By presenting clear, live updates in a simple format, the application helps users save time, reduce frustration, and make confident decisions without guesswork.

What Makes Our Product Unique?

Vibe Check differentiates itself through its location-locked posting system, which ensures that notes can only be submitted and viewed within a defined geographic radius. This increases reliability and prevents irrelevant or misleading content. Additionally, preset expiration times automatically remove outdated notes, maintaining data freshness. A live heatmap further enhances visibility by showing overall activity levels in an area at a glance.

Future Potential

Looking ahead, Vibe Check has opportunities for business partnerships with cafés, venues, and local establishments that may benefit from real-time engagement insights. Gamification features such as badges and contribution streaks could encourage consistent user participation. The platform may also introduce personalized notifications and trend summaries to enhance user experience and long-term engagement.

Main Use Cases

Checking a Busy Restaurant Before Committing

Actors: John (Registered User), Vibe Check (System)

Assumptions:

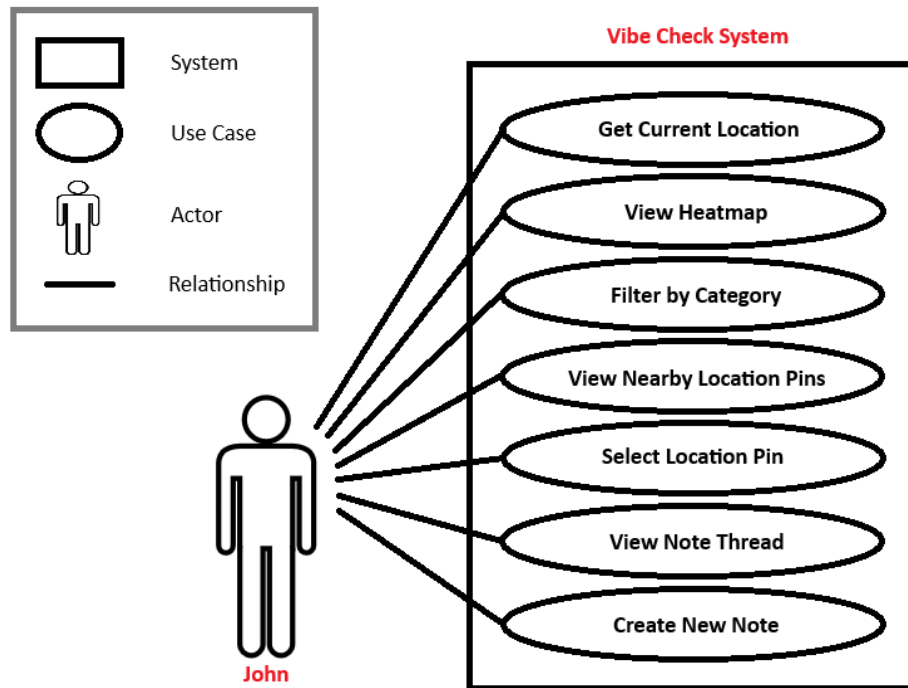
- John has access to an Internet connected device
- John has location services enabled on his device
- There are existing notes in the area created recently

Use Case: 1

John is walking around his neighborhood and is looking for a place to eat. He has to be back at work soon and does not want to waste time walking to a restaurant only to find out that there is a long wait. He opens Vibe Check and sees a heatmap of the neighborhood that he is in. He filters the heatmap by 'Restaurants' and selects a restaurant that is of interest to him. He taps on the restaurant pin and reads a recent note that says, "The line is going around the block! It's about a 30 minute wait just to get inside." John reads the replies to this note and sees other users confirming the update. John then taps on another restaurant that has a note saying, "Quick service, got great food in less than 15 minutes." John also notices a few people reacted positively to this note. John feels more confident in making the decision to go to the second restaurant and in being able to get back to work on time because the information is current and from people who were actually at that location recently. After eating, John creates his own note on the restaurant in Vibe Check that reflects his experience, allowing future users to also get up to date information on the location.

Benefits for John:

- John can make an informed decision on how and where to spend his time without any guesswork.
- John can get a crowdsourced opinion on a location he has not been to before.



Finding a Quiet Study Spot on Campus

Actors:

Monica (Registered User, University Student), Vibe Check (System)

Assumptions:

- Monica has access to an Internet connected device
- Monica has location services enabled on her device
- There are existing notes in the campus area created recently

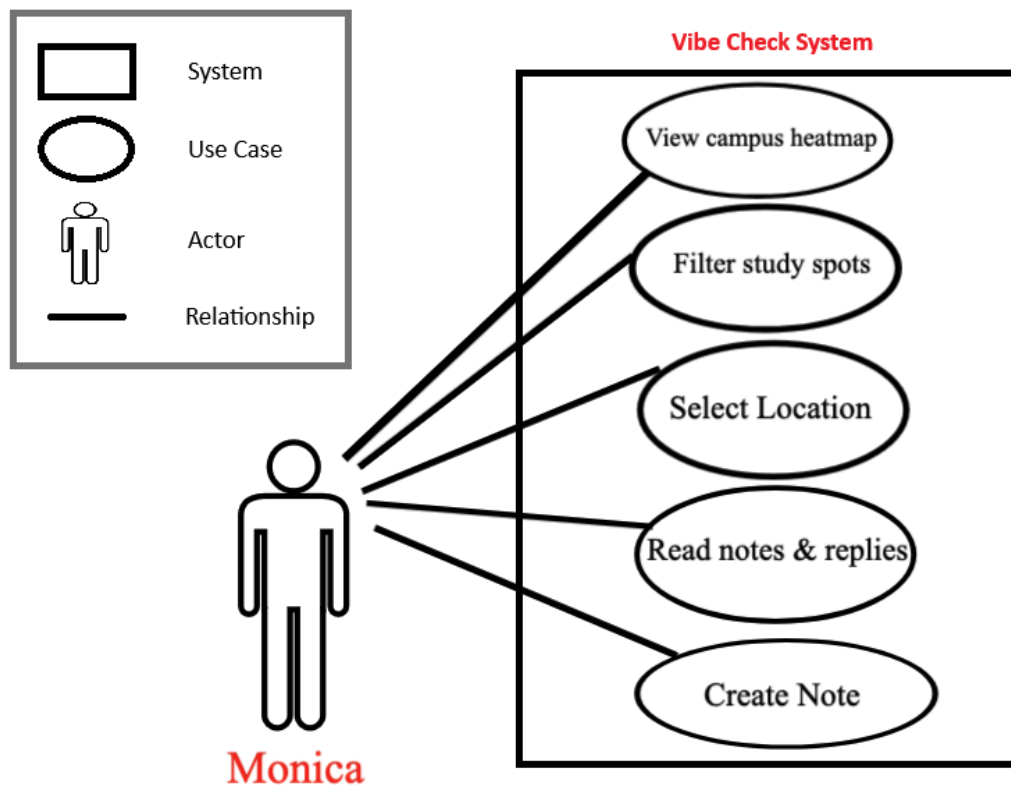
Use Case: 2

Monica is a university student preparing for her midterm exams & needs a quiet place to study for a few hours. She doesn't want to walk to the library only to find it's packed and noisy. Monica opens Vibe check & views the heatmap of her campus area. She filters the heatmap by 'Study spots' and notices several locations with varying levels of activity. She taps on the main library and reads a note posted 10 minutes ago that says, "Every seat is taken & it's pretty loud here with group projects going on". Monica checks the replies & sees multiple users confirming the crowded conditions. She then selects a smaller study lounge in another building & reads a recent note: "Super quiet right now, plenty of empty tables by the windows." The note has several positive reactions from other students. Confident in her choice, Monica heads to the study lounge & successfully finds a peaceful spot to study. After settling in, she

creates her own note on Vibe Check confirming the quiet atmosphere and available seating, helping other students who might be searching for study spots later.

Benefits for Monica:

- Monica saves time by avoiding crowded or unsuitable study locations
- Monica can find the ideal environment for her current needs (quiet, available seating)
- Monica contributes real-time information that helps fellow students make better decisions



Reviewing Places by Popularity

Actors: Lex (Registered User, Worker), Vibe Check (System)

Assumptions:

- Lex has access to the internet with a mobile device
- Lex has location enabled services turned on
- There is existing reviews and comments on location

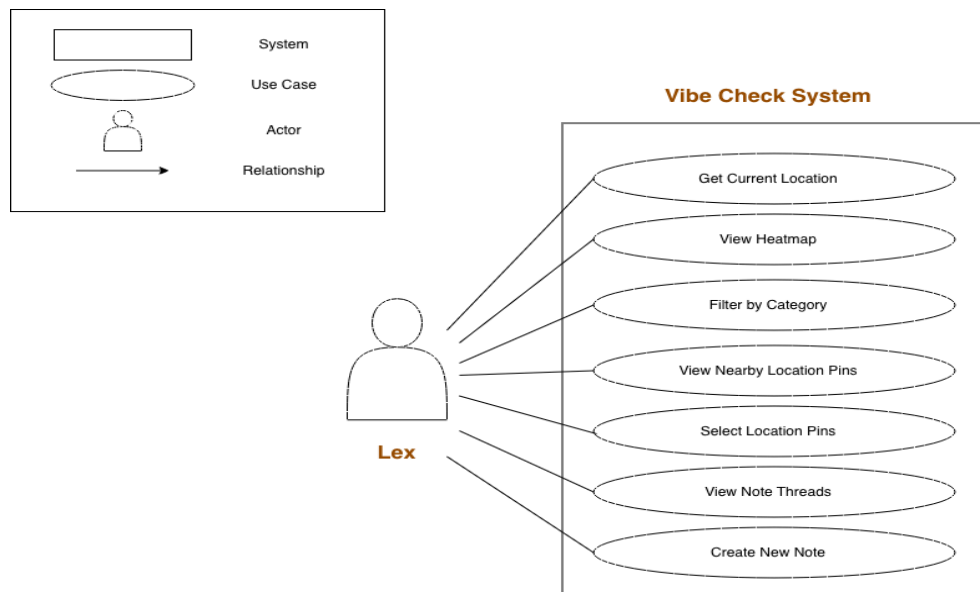
Use Case: 3

Lex is a hybrid worker, but prefers to work in a more work oriented environment during her times she is not in the office. She looks for places to work, but is unable to find something that she is sure about which

matches the environment she is looking for. She finds an app called Vibe Check. From the app, she is able to get a better insight on where she goes and the aesthetic it goes for under a tab of quiet places. Lex is happy to see the specificity of vibes of a place through the check-in boards, as well as the history of recent activities at a certain place. After reviewing multiple locations' look and check-in boards comments, she sees a perfect spot with a photo attached and a comment saying "I found this hidden spot nearby my college campus, it's really quiet but very aesthetically pleasing to work in". After visiting the location she is pleased with her experience. She reacts to the other person's comment and shares it with the community for people who are looking for a similar experience.

Benefits for Lex:

- Lex is able to find quiet places around her to focus on her work
- Lex is able to find an environment that inhibits her work style
- Lex saves time in finding places instead of visiting places that might have unavailable seating



Notified of Nearby Trendy Location

Actors:

Chris (Registered User), Notification

Assumptions:

- Chris has access to the internet with a mobile device
- Chris has location enabled services turned on

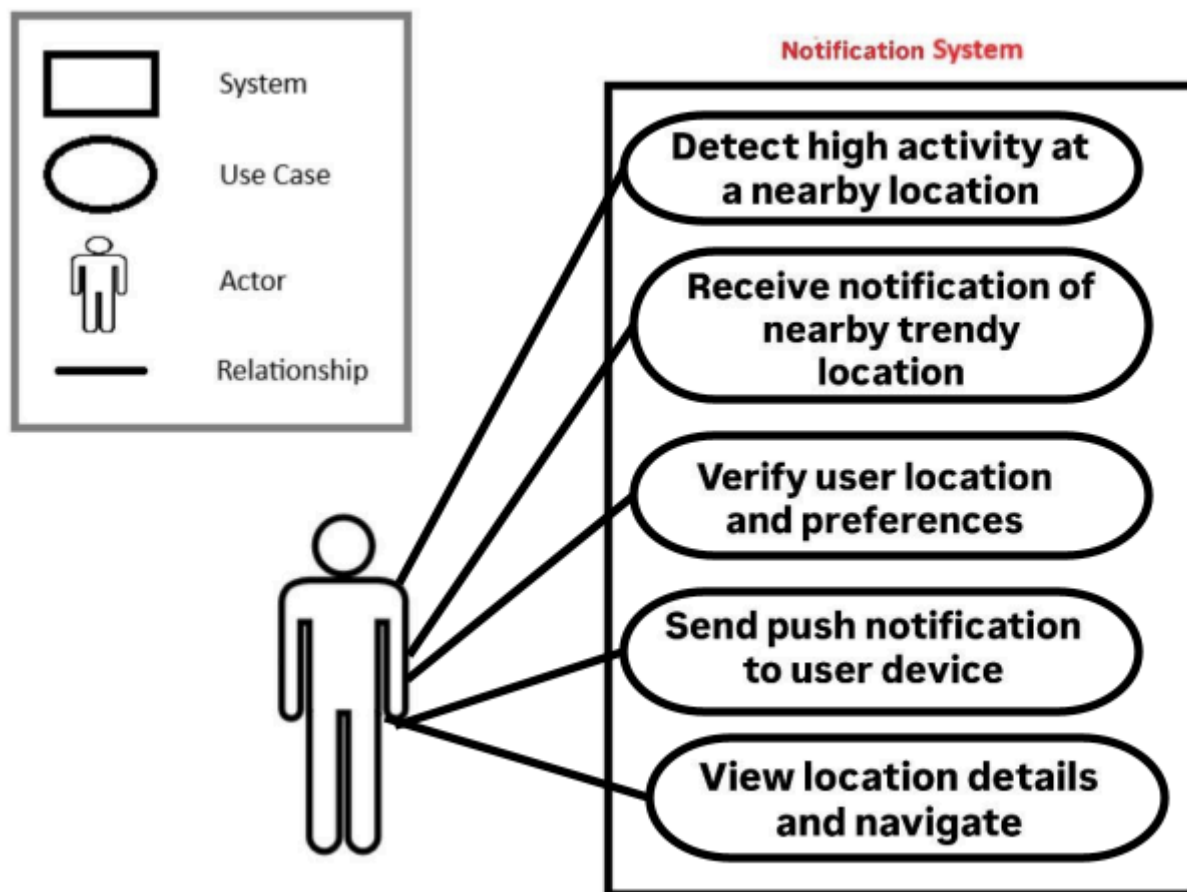
Use Case: 4

Chris is at his house trying to make Friday night plans, he likes to make new friends so when he goes out he prefers to go somewhere with a lot of people to meet. He gets a notification from the Vibe Check app.

The notification tells him that a venue near him is bustling tonight. Chris is happy to have quickly found a high energy spot to spend his evening.

Benefits for Chris:

- Chris is able to find a lively place to meet new people
- Chris saves time looking for places or showing up to places that are dead



Reviewing Personal Actors: John (Registered User, Business Owner), Vibe Check (System)

Assumptions:

- John claimed his business listing on Vibe Check
- John has access to the app
- There is existing reviews and comments on location

Use Case: 5

Carlos owns a mid-sized coffee shop and wants to understand how customers are perceiving the atmosphere during different times of the day. He monitors his location on Vibe Check throughout the day. During the rushes he is able to read the notes/comments of his cafe and reads notes that say “great music

at low volumes, and pretty comforting place to study at”. He also is able to read other notes at certain times saying “It was too busy and loud during our visit”. He lowers his volume and increases seating at more busier hours due to the feedback and optimizes the way he works through crowdsourced comments

Benefits for John:

- Carlos is able to receive feedback and constructive criticism on his business.
- Carlos is able to make adjustments to customer sentiment.
- Carlos builds a stronger reputation catering to customers over time.

Reviewing Places by Popularity

Actors: Abel (Registered User, Community Contributor), Vibe Check (System)

Assumptions:

- Abel has access to the internet with a mobile device
- Abel has location enabled services turned on
- There is an accessible bulletin board/comment section to advertise

Use Case: 6

- Abel wants to support his community event at a local care to bring in new members, but has trouble advertising it. He remembers that Vibe Check had a comment section and posts real-time that “I am hosting a small mixer here if anybody wants to talk and make new friends”.

Benefits for John:

-

Main Data Items and Entities

User

A user is a person who uses the application to view nearby locations and interact with content. A User can create Notes and can reply to or react to existing Notes.

Location

A Location represents a physical place shown on the map (e.g., café, library, venue, park). Notes are tied to a specific Location to ensure that updates are relevant to that place.

Note

A Note is a short, location-locked post created by a User to describe what is happening at a Location in real time. Each Note has a timestamp and an expiration time so that outdated information is automatically removed.

Category Tag

A Category Tag is a label used to classify a Note or Location (e.g., restaurant, study spot, venue line, parking). Category Tags allow users to filter and view relevant content more efficiently.

Reply

A Reply is a response to a specific Note that provides additional context or updated information. Each Reply belongs to one Note.

Reaction

A Reaction represents a quick feedback action (such as an upvote or similar interaction) that a User can apply to a Note. Reactions help indicate which notes are useful or accurate.

Heatmap Activity

Heatmap Activity is a visual representation of recent activity in a geographic area. It is generated from recent Notes and user interactions within a specific time window and helps users quickly assess how active an area is.

User Profile

A user profile stores user preferences such as notification settings, default radius, preferred categories. It tracks user contribution stats (for future gamification).

Notification

A notification represents a system notification (, trending location nearby, etc.)

Logical Relationships

- A User can create many Notes.
- A Note belongs to one Location.
- A Note can have many Replies.
- A User can create many Replies.
- A User can create many Reactions, and each Reaction applies to one Note.
- A Location can have many Notes over time.
- Heatmap Activity is derived from recent Notes and interactions.
- Each User has one Profile
- Profile stores preferences and activity history
- A Notification belongs to one User

- A Notification references one Note or Location

Functional Requirements

Authentication

- FR-01: The user shall be able to create an account with an email and password.
- FR-02: The user shall be able to log in and log out.
- FR-03: The user shall be required to be authenticated in order to create notes, replies, or reactions.

Location

- FR-04: The user shall be able to grant permission for the app to access their current location.
- FR-05: The user shall be able to view nearby locations on the map.
- FR-06: The user shall be able to select a specific location from the map.

Heatmap

- FR-07: The user shall be able to view a heatmap showing activity in the nearby area.
- FR-08: The user shall be able to filter the heatmap by category.

Notes

- FR-09: The user shall be able to view notes related to a specific location.
- FR-10: The user shall be able to reply to notes posted by other users.
- FR-11: The user shall be able to react to notes posted by other users.
- FR-12: The user shall be able to post a note at a specific location when they are within a defined radius.

Notification

- FR-13: The user shall be able to receive a notification when a trendy location is nearby.
- FR-14: The user shall be able to enable or disable notifications in settings.

Search/Filter

- FR-15: The user shall be able to search for locations by name or category.

Non Functional Requirements

Performance

- NFR-01: The system shall load the heatmap of the user's vicinity within 3 seconds.

Reliability

- NFR-02: The system shall keep notes, replies, and reactions persistent until their preset expiration time.

Security

- NFR-03: The system shall securely store user passwords using hashing and salting.
- NFR-04: The system shall not expose the user's exact location to other users.

Usability

- NFR-05: The system shall warn the user of missing or denied location access and offer a fallback solution.

Scalability

- NFR-06: The system shall support an increasing number of notes as the user base grows.

Compliance

- NFR-07: The system shall request user permission before accessing location services.

Maintainability

- NFR-08: The system shall follow a consistent coding standard and project structure across the frontend and backend.

Competitive Analysis:

Competitors	Comparison
Waitz (Occuspace)	Waitz uses passive sensor technology (Bluetooth signals) to measure real-time busyness at campus locations. While effective within its network, it requires hardware installation and institutional agreements to operate. Vibe Check requires no sensors or university buy-in any user can contribute a note from any location, making it far more scalable and accessible beyond campus boundaries."
Google Maps	Google Maps "Popular Times" feature relies on historical trends to estimate how busy a location might be at a given time. While Vibe Check uses Heatmap Activity to get live updates for a given location. This allows users to get more accurate information about a place before they go.
Yelp	Yelp relies on past reviews to get a sense of what the experience of a place may be, however these can quickly become outdated. Vibe Check allows users

	to leave notes that expire after a given amount of time, this keeps reviews timely and accurate.
Snapchat Snap Map	Snapchat Snap Map shows where people are hanging out in real time, but the content is unstructured (random snaps) and doesn't give you a clear "vibe" signal about a place.
Swarm by Foursquare	uses check-ins to imply presence at a location, but doesn't convey atmosphere or crowding in a useful way.

Competitive Analysis Comparison Table

Key comparison area	Google Maps	Yelp	Snapchat Snap Map	Swarm (Foursquare)	Waitz (Occuspace)	Vibe check
Core value: real-time “vibe” decisioning (quiet/loud/crowded/active)	Medium (busyness, not “why”)	Medium (reviews lag)	Low (unstructured)	Low (check-ins)	Medium (busyness only)	Strong, purpose-built
Freshness control (keeps info current)	Medium (live busyness sometimes)	Low (reviews persist)	Low (not freshness-driven)	Low	Medium (sensor-based, no expiry)	Strong (expiration)
Trust & relevance (credible for this exact place)	Medium (algorithmic)	Medium (varies by reviewer)	Low (context unclear)	Medium (presence implied)	Medium (sensor-based)	Strong (location-locked + confirmations)
Discovery UX (scan area fast)	Strong (map + popular times)	Medium (search/reviews)	Strong (map browsing)	Medium	Low (campus only)	Strong (heatmap + filters)
Content density & coverage (works everywhere)	Very High	High	Very High	Low-Med	Low	Low early (cold start)
Distribution	Very High	High	Very High	Low-Med	Low	Low (new)

& habit (default app users open)						
Operational risk (spam/trolling /moderation)	Medium	Medium	High	Medium	Low	High